

Chapter – 2

Purple E-Commerce Database

2.1 Introduction

The entities required for the Purple E-Commerce Database are identified based on the business requirements of an online beauty and personal care shopping platform. The attributes of each entity, along with their primary keys, foreign keys, unique keys, and attribute types, are determined to ensure efficient data organization. The relationships among the entities are identified to represent customer interactions, product management, order processing, payments, deliveries, and inventory management. This analysis forms the foundation for designing the Entity Relationship (ER) Diagram, relational schema, and database implementation while ensuring data integrity, consistency, and minimum redundancy.

2.2 Entity Analysis

2.2.1 Entity: Customer

Attribute	Key Type
Customer_ID	Primary Key
Customer_Name	Not Null
Email	Unique Key
Phone_Number	Unique Key
Address	Not Null

2.2.2 Entity: Category

Attribute	Key Type
Category_ID	Primary Key
Category_Name	Unique Key

2.2.3 Entity: Product

Attribute	Key Type
Product_ID	Primary Key
Product_Name	Not Null
Brand	Not Null
Price	Not Null
Stock	Not Null

Category_ID	Foreign Key
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2.2.4 Entity: Order

Attribute	Key Type
Order_ID	Primary Key
Customer_ID	Foreign Key
Order_Date	Not Null
Total_Amount	Not Null
Order_Status	Not Null

2.2.5 Entity: Order_Item

Attribute	Key Type
Order_Item_ID	Primary Key
Order_ID	Foreign Key
Product_ID	Foreign Key
Quantity	Not Null
Price	Not Null

2.2.6 Entity: Payment

Attribute	Key Type
Payment_ID	Primary Key
Order_ID	Foreign Key
Payment_Method	Not Null
Payment_Status	Not Null

2.2.7 Entity: Delivery

Attribute	Key Type
Delivery_ID	Primary Key
Order_ID	Foreign Key
Tracking_Number	Unique Key
Delivery_Status	Not Null

2.2.8 Entity: Supplier

Attribute	Key Type
Supplier_ID	Primary Key
Supplier_Name	Not Null
Contact_Number	Unique Key
Email	Unique Key

2.2.9 Entity: Admin

Attribute	Key Type
Admin_ID	Primary Key
Admin_Name	Not Null
Email	Unique Key
Password	Not Null

2.3 Entity Relationships

Entity 1	Relation	Entity 2	Cardinality	Participation
Customer	Places	Order	One-to-Many	Partial / Total
Category	Contains	Product	One-to-Many	Total / Total
Order	Includes	Order_Item	One-to-Many	Total / Total
Product	Appears In	Order_Item	One-to-Many	Partial / Total
Order	Has	Payment	One-to-One	Total / Total
Order	Has	Delivery	One-to-One	Total / Total
Supplier	Supplies	Product	One-to-Many	Partial / Total
Admin	Manages	Product	One-to-Many	Partial / Total
Admin	Manages	Order	One-to-Many	Partial / Total

2.4 Conclusion

The entity analysis for the Purple E-Commerce Database successfully identifies the essential entities, their attributes, primary keys, foreign keys, and relationships required to support an online beauty and personal care shopping platform. The analysis ensures efficient management of customers, products, categories, suppliers, orders, payments, deliveries, and administrative activities. By defining appropriate relationships, cardinality, and participation constraints, the database minimizes redundancy while maintaining data integrity and consistency. This entity analysis serves as the foundation for developing the Entity Relationship (ER) Diagram, relational schema, normalization process, and the overall implementation of the Purple E-Commerce Database Management System.